

Hootstuf 3

Lesing 3.4

①

Recap:
~~nodal~~
(Nodal analysis)

$$\boxed{\text{KCL}} \rightarrow \sum I_{\text{OUT}} = \sum I_{\text{IN}}$$

↳ Node analyse

- * Skryf strome in node-spannings en R'e.
Gebruik Ω -se-Wet.

$$i = \frac{V_{\text{FROM}} - V_{\text{TO}}}{R} = \frac{+V_R}{R}$$

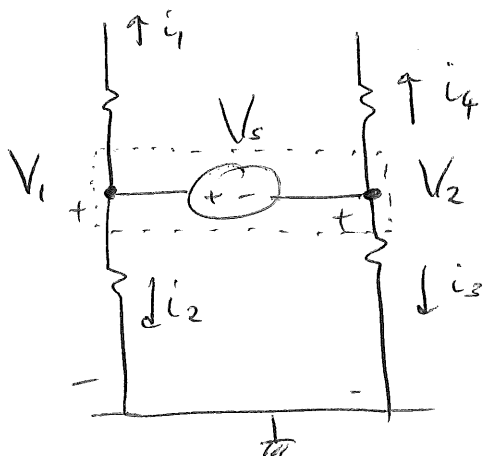
- * V_R is die spanning oor die weerstand.

$$\begin{array}{c} + \quad V_R \quad - \\ \xrightarrow{i} \end{array}$$

→ Supernodesse:

(Spanningsbronne)

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KVL: * $-V_1 + V_s + V_2 = 0$
random supernode

KCL: * $i_1 + i_2 + i_3 + i_4 = 0$
by supernode

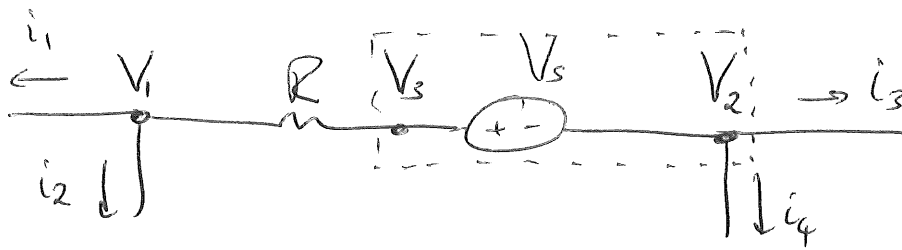
- * Nie noodlik om stroom deur V_s in Ω -Wet te skryf nie (Ω -Wet slegs vir R'e!)

- * Dus, vorm geslote grens rondom bron.

- * Sommer strome by supernode.

- * Skryf V_s in V_1 en V_2 .

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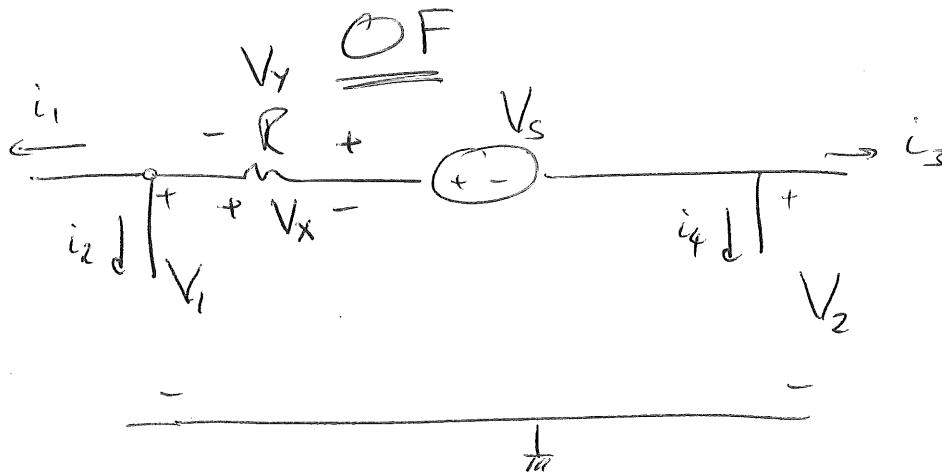
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* Vorm gesloten grens rondom V_s .

Node 1: $i_1 + i_2 + \frac{V_1 - V_3}{R} = 0$

(Supernode:) Node 2: $i_3 + i_4 + \frac{V_3 - V_1}{R} = 0$

Branch: $V_s = V_3 - V_2$.



Node 1: $i_1 + i_2 + \frac{V_1 - V_s - V_2}{R} = 0$

$V_x = V_1 - V_s - V_2$ is spanning oor R :

⌚ KVL: $-V_1 + V_x + V_s - V_2 = 0$

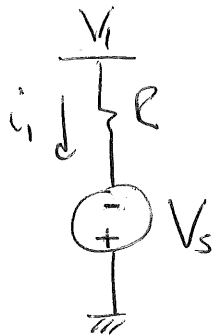
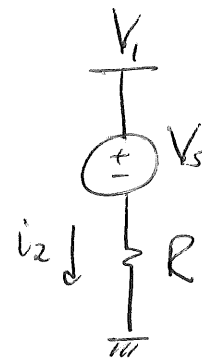
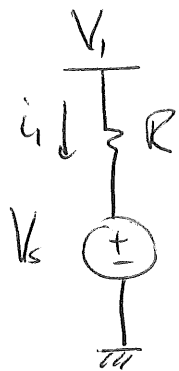
$V_x = V_1 - V_s - V_2$.

Node 2: $i_3 + i_4 + \frac{V_2 - (-V_s) - V_1}{R} = 0$

⌚ KVL: $-V_2 - V_s + V_y + V_1 = 0$

$$V_y = V_2 + V_s - V_1$$

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$$i_1 =$$

$$i_2 =$$

Recap
(Mesh analysis)

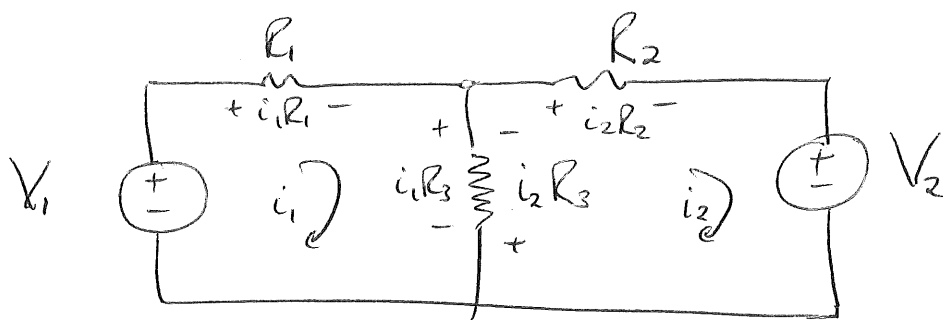
KVL

$$\rightarrow \sum V_{\text{loop}} = 0$$

(4)

↳ Mesh analysis
(Loop analysis)

* Skript spannings its R-Net
(Onton teken conversie!)



$$\sum i_{\text{loop}} R = 0.$$

$$* -V_1 + i_1 R_1 + i_1 R_3 - i_2 R_3 = 0$$

$$i_1 (R_1 + R_3) - i_2 R_3 = V_1$$

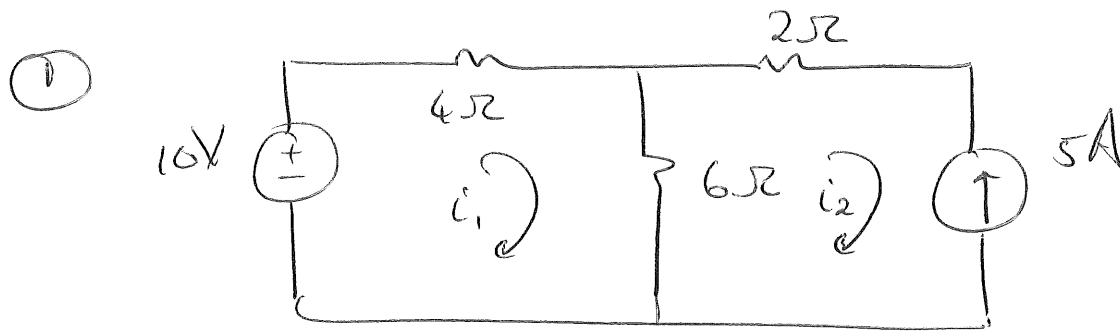
$$* -i_1 R_3 + i_2 R_3 + i_2 R_2 + V_2 = 0$$

$$-i_1 R_3 + i_2 (R_2 + R_3) = -V_2.$$

$$\Rightarrow \begin{bmatrix} R_1 + R_3 & -R_3 \\ -R_3 & R_2 + R_3 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ -V_2 \end{bmatrix}$$

* Lus analyse met stroombronnen.

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$$i_2 = -5A$$

* i_2 is enigste lusstroom waar 5A bron vloeit.

∴ KVL: $-10 + i_1 \times 4 + i_1 \times 6 - i_2 \times 6 = 0$

$$10i_1 - 6i_2 = 10$$

$$10i_1 - 6(-5) = 10$$

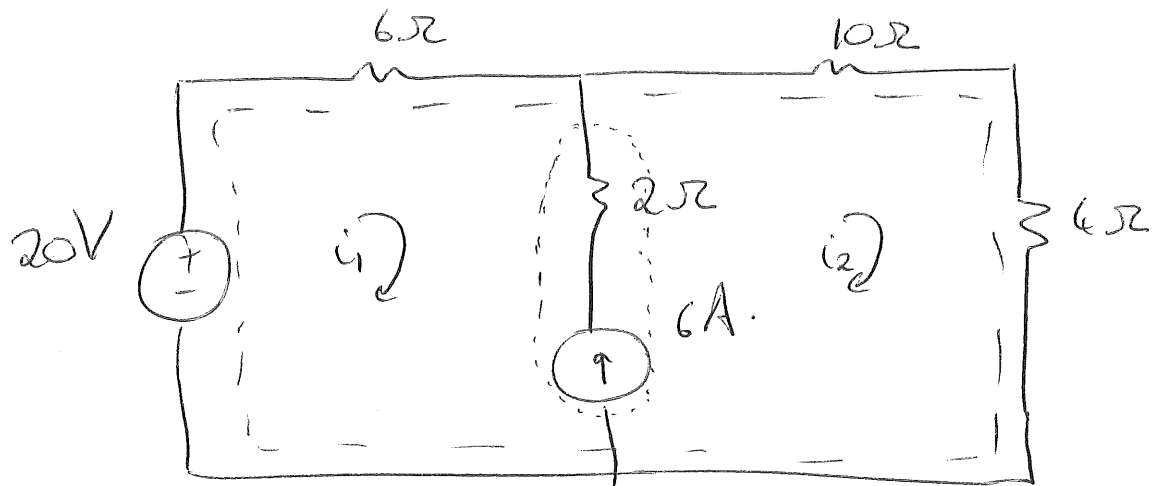
$$10i_1 = 10 - 30$$

$$i_1 = -2A \rightarrow$$

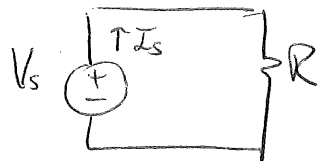
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Supermeshes / Superlusse

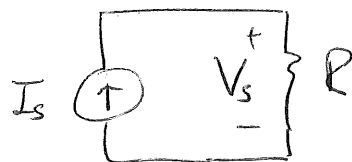
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* Net soos daar stroom vanuit V-bron ulei,



is daar spanning oor I-bron:



* Maar spanning oor I-bron kan nie geskryf word its Ω -Wet nie.

* Superlus: vorm spanningslus rondom I-bron en enige elemente in serie met bron.

(Kombineer i_1 en i_2).

→ Superlus: $-20 + 6i_1 + 10i_2 + 4i_2 = 0$.

→ KCL: $i_1 + 6 = i_2$ (Boonste node).

Neem kennis:

Supernodusse:

Vergelyking 1: Som van strome vir geslote grens.

Vergelyking 2: Bestrywing van V-bron ito node-spannings.

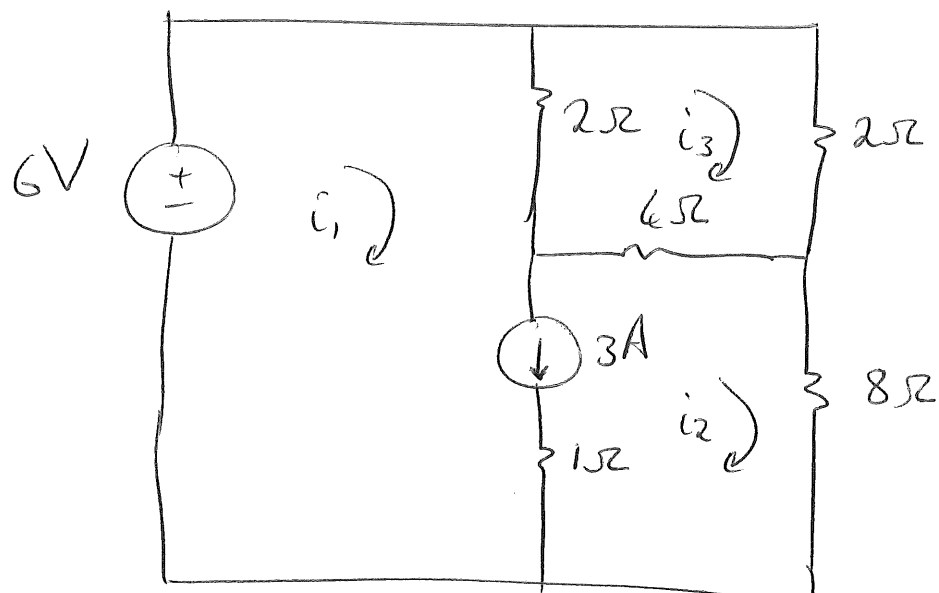
Superlus

Vergelyking 1: Som van spannings rondom superlus.
(Som van spannings rondom gekombineerde lus)

Vergelyking 2: Bestrywing van I-bron ito lusstrome.

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V6



V6

